Air Quality Analysis and AQI Prediction Using Machine Learning

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1. Abstract

Air pollution is a major environmental and public health issue, particularly in urban regions such as Delhi. This project focuses on analyzing air quality data and predicting the Air Quality Index (AQI) using statistical and machine learning techniques. The dataset includes pollutant concentrations such as PM2.5, PM10, and NO_2 along with AQI values. Data preprocessing techniques were applied to handle missing values and prepare the dataset for analysis. Exploratory Data Analysis (EDA) was conducted to understand pollution trends and relationships between pollutants. A Linear Regression model was used to predict AQI based on pollutant concentrations, while an ARIMA time-series model was applied to capture temporal patterns in AQI data. The results show strong correlations between particulate matter levels and AQI, highlighting the importance of controlling emissions to improve urban air quality.

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2. Introduction

Air pollution has become one of the most critical environmental challenges affecting human health and urban sustainability. Rapid industrialization, vehicular emissions, and population growth have significantly contributed to increasing pollution levels in major cities worldwide.

The Air Quality Index (AQI) is a standardized metric used to measure and communicate air pollution levels. It considers several pollutants such as particulate matter (PM2.5 and PM10), nitrogen dioxide (NO_2), and others.

This project aims to analyze AQI data for Delhi and develop predictive models that can estimate AQI values based on pollutant concentrations. By applying machine learning and time-series analysis techniques, the study provides insights into pollution trends and potential forecasting methods.

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3. Objectives

The primary objectives of this project are:

- To analyze air pollution data using statistical and visualization techniques
- To identify relationships between pollutants and AQI
- To develop a regression model for predicting AQI

- To apply time-series forecasting to understand pollution trends
- To provide insights that may support environmental monitoring and decision-making

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4. Dataset Description

The dataset used in this project contains historical air quality measurements from Delhi. It includes pollutant concentration levels and AQI values recorded over time.

Key features in the dataset include:

Feature	Description
Date	Date of observation
PM2.5	Fine particulate matter concentration
PM10	Coarse particulate matter concentration
NO_2	Nitrogen dioxide concentration
AQI	Air Quality Index value

Data preprocessing steps included:

- Handling missing values using forward fill technique
- Converting date columns into datetime format
- Preparing the dataset for time-series analysis

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5. Data Preprocessing

Data preprocessing is an essential step in machine learning workflows. The dataset was first inspected for missing values and inconsistencies.

Steps performed:

1. Loading the dataset using Pandas
2. Checking for null values
3. Handling missing values using forward fill
4. Converting the date column into datetime format
5. Preparing the dataset for visualization and modeling

These steps ensured that the dataset was clean and suitable for analysis.

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6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand patterns and relationships in the dataset.

AQI Trend Analysis

A time-series plot was created to observe AQI fluctuations over time. This visualization helps identify pollution peaks and seasonal patterns.

Monthly Pollution Patterns

The dataset was grouped by month to calculate average AQI values. This analysis reveals seasonal pollution trends and identifies months with higher pollution levels.

Pollutant Relationships

A scatter plot was used to analyze the relationship between PM2.5 and PM10 concentrations. The visualization shows that higher PM2.5 levels are generally associated with higher PM10 levels.

Correlation Analysis

A correlation heatmap was generated to examine relationships between all variables in the dataset. Strong correlations indicate pollutants that significantly influence AQI values.

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7. Machine Learning Model – Linear Regression

A Linear Regression model was used to predict AQI based on pollutant concentrations.

The regression model follows this equation:

$y = \beta 0 + \beta 1x1 + \beta 2x2 + \beta 3x3$

Where:

- **y** represents the predicted AQI value
- **x1, x2, x3** represent pollutant concentrations (PM2.5, PM10, NO_2)
- **β coefficients** represent the influence of each pollutant on AQI

The model was trained using the Scikit-learn library and evaluated using the R^2 score.

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8. Time Series Forecasting Using ARIMA

To analyze how AQI changes over time, a time-series forecasting model was applied.

The ARIMA model captures temporal dependencies in sequential data.

Its mathematical representation is:

$$y_t = c + \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \dots + \varphi_p y_{t-p} + \varepsilon_t$$

Where:

- $\mathbf{y_t}$ represents AQI at time t
- $\boldsymbol{\varphi}$ represents autoregressive coefficients
- $\boldsymbol{\varepsilon_t}$ represents the error term

The ARIMA model helps identify patterns in AQI trends and provides a basis for forecasting future pollution levels.

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9. Results and Insights

The analysis provided several key insights:

- PM2.5 and PM10 concentrations strongly influence AQI levels.
- Seasonal trends indicate that pollution levels increase during certain months.
- The regression model successfully predicts AQI values using pollutant concentrations.
- Time-series analysis captures temporal patterns in pollution data.

These findings demonstrate the importance of monitoring particulate matter emissions to control air pollution.

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10. Conclusion

This project analyzed air quality data using data science and machine learning techniques. Data preprocessing and exploratory analysis revealed strong relationships between pollutant concentrations and AQI levels.

The Linear Regression model demonstrated the ability to predict AQI based on pollutant concentrations, while the ARIMA model captured temporal trends in air pollution data.

Overall, the study highlights the usefulness of machine learning techniques in environmental monitoring and pollution analysis.

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11. Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Statsmodels

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12. Future Improvements

Future work may include:

- Using advanced machine learning models such as Random Forest or XGBoost
- Incorporating weather data for improved predictions
- Developing real-time AQI prediction systems
- Creating interactive dashboards for pollution monitorin